

Network Analysis		Semester	3
CourseCode	BEC304	CIEMarks	50
TeachingHours/Week (L:T:P: S)	3:0:0	SEE Marks	50
TotalHoursofPedagogy	40	TotalMarks	100
Credits	03	Exam Hours	3
Examinationtype (SEE)	Theory		
Courseobjectives: 1. Applymeshand nodal techniquesto solve an electrical network. 2. SolvedifferentproblemsrelatedtoElectricalcircuitsusingNetworkTheoremsandTwo port network. 3. Familiarizewiththeuseof Laplacetransformstosolvenetworkproblems. 4. Studytwo portnetworkparametersandtheir applications. 5. StudyofRLC Seriesandparalleltunedcircuit.			
Teaching-LearningProcess(GeneralInstructions) ThesearesampleStrategies,whichteacherscanusetoacceleratetheattainmentofthevarious course outcomes. ThesearesampleStrategies,whichteachercanusetoacceleratetheattainmentofthevarious course outcomes. Lecturemethod(L)doesnotmeanonlytraditionallecturemethod,butdifferenttypeof teaching methods may be adopted to develop the outcomes. Encouragecollaborative(Group)Learninginthe class. AskatleastthreeHOTS(HigherorderThinking)questionsinthe class,which promotes critical thinking. AdoptProblemBasedLearning(PBL),whichfostersstudents' Analyticalskills,develop thinking skills such as the ability to evaluate, generalize, and analyze information rather than simply recall it. Topicswill beintroduced ina multiplerepresentation. Showthedifferentwaystosolvethesameproblemandencouragethestudentsto come up with their own creative ways to solve them. Discusshoweveryconceptcanbeappliedtothe realworld-andwhenthat'spossible, it helps improve the students' understanding. AdoptFlippedclasstechniquebysharingthematerials/SampleVideospriortothe class and have discussions on the topic in the succeeding classes.			
Module-1			
BasicConcepts: Practicalsources,Sourcetransformations,NetworkreductionusingStar-Deltatransformation,Loopandnodeanalysiswithlinearlydependentandindependent sources for DC and AC networks.			
Module-2			
NetworkTheorems: Superposition,Millman's theorems,Thevenin's andNorton's theorems, Maximum Power transfer theorem.			
Module-3			

Transient behavior and initial conditions: Behavior of circuit elements under switching condition and their Representation, evaluation of initial and final conditions in RL, RC and RLC circuits for AC and DC excitations.
Module-4
Laplace Transformation & Applications: Solution of networks, step, ramp and impulse responses, waveform Synthesis.
Module-5
Two port network parameters: Definition of Z, Y, h and Transmission parameters, modelling with these parameters, relationship between parameters sets. Resonance: Series Resonance: Variation of Current and Voltage with Frequency, Selectivity and Bandwidth, Q-Factor, Circuit Magnification Factor, Selectivity with Variable Capacitance, Selectivity with Variable Inductance. Parallel Resonance: Selectivity and Bandwidth, Maximum Impedance Conditions with C, L and f Variable, current in Anti-Resonant Circuit, The General Case-Resistance Present in both Branches.
Course outcome (Course Skill Set) At the end of the course, the student will be able to : 1. Determine currents and voltages using source transformation/source shifting/mesh/ nodal analysis and reduce given network using star- delta transformation. 2. Solve problems by applying Network Theorems and electrical laws to reduce circuit complexities and to arrive at feasible solutions. 3. Analyse the circuit parameters during switching transients and apply Laplace transform to solve the given network 4. Evaluate the frequency response for resonant circuits and the network parameters for two port networks

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). The student is declared as a pass in the course if he/she secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- There are 25 marks for the CIE's Assignment component and 25 for the Internal Assessment Test component.
- Each test shall be conducted for 25 marks. The first test will be administered after 40-50% of the coverage of the syllabus, and the second test will be administered after 85-90% of the coverage of the syllabus. The average of the two tests shall be scaled down to 25 marks
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The schedule for assignments shall be planned properly by the course teacher. The teachers should not conduct two assignments at the end of the semester if two assignments are planned. Each assignment shall be conducted for 25 marks. (If two assignments are conducted then the sum of the two assignments shall be scaled down to 25 marks)
- The final CIE marks of the course out of 50 will be the sum of the scale-down marks of tests and assignment/s marks.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**).

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.

Suggested Learning Resources:**Books**

1. M.E. Van Valkenburg (2000), Network Analysis, Prentice Hall of India, 3rd edition, 2000, ISBN: 9780136110958.
2. Roy Choudhury- Networks and Systems, 2nd edition, New Age International Publications, 2006, ISBN: 9788122427677

Reference Books:

3. Hayt, Kemmerly and Durbin- Engineering Circuit Analysis, TMH 7th Edition, 2010.
4. J. David Irwin/R. Mark Nelms- Basic Engineering Circuit Analysis John Wiley, 8th ed, 2006.
5. Charles K Alexander and Mathew N O Sadiqu- Fundamental of Electric Circuits, Tata McGraw-Hill, 3rd ed, 2009.